Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 1 / 方法 1 [NT04-M01]

A residue cycle must be proved at the correct state level. Multiplicative order handles units; prime-power valuations handle nonunits and lifting. Separate these roles before combining them by CRT.

剩余周期须在正确的状态层面证明。乘法阶处理单位，质数幂赋值处理不可逆部分与提升；先区分这些作用，再用中国剩余定理合并。

Certify the multiplicative order before reducing exponents

约化指数前先确认乘法阶

Procedure

1. For $\gcd(a,m)=1$, define the order as the least positive d with $a^d \equiv 1 \pmod{m}$.

2. Find a candidate exponent from a known group bound or direct calculation. For each prime $q \mid d$, verify $a^{d/q} \not\equiv 1$ to certify exactness.

3. Then $a^n = 1$ precisely when $d \mid n$; a prescribed minus-one power also constrains the two-part of d.

4. For prime moduli, use $d \mid p-1$ to restrict prime candidates, and verify the surviving prime.

通用步骤

1. 当 $\gcd(a,m)=1$，乘法阶为使 $a^d \equiv 1 \pmod{m}$ 的最小正整数 d。

2. 由群阶界或直接计算找候选指数。对每个质因数 $q \mid d$，检验 $a^{d/q} \not\equiv 1$，以确认准确阶数。

3. 此后 $a^n = 1$ 恰当 $d \mid n$；指定幂为负一还会限制 d 的二因子部分。

4. 质数模下以 $d \mid p-1$ 限制质数候选，再验证存活质数。

> Example. Find the order of 3 modulo 13 and decide whether $13 \mid 3^{100}+1$.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 1 / 方法 1 [NT04-M01] (continued / 续)

> 例题：求 3 模 13 的乘法阶，并判断 $13 \mid 3^{100}+1$ 是否成立。

Worked solution

1. $3^3 = 27 \equiv 1 \pmod{13}$, while $3 \not\equiv 1$, so the order is exactly 3.

2. Since $100 \equiv 1 \pmod{3}$, one has $3^{100} \equiv 3 \pmod{13}$.

3. The expression is congruent to 4, so divisibility fails. A power of odd order cannot equal -1 modulo an odd prime.

例题解析

1. $3^3 = 27 \equiv 1 \pmod{13}$，而 $3 \not\equiv 1$，故阶恰为 3。

2. 因 $100 \equiv 1 \pmod{3}$，有 $3^{100} \equiv 3 \pmod{13}$。

3. 原式余 4，故不能整除；在奇质数模下，奇数阶的幂也不可能等于 -1。

Euler gives an exponent that works; it need not be the least one.

欧拉定理给出可行指数，但未必是最小指数。

If $a^m \equiv -1 \pmod{p}$ for odd prime p, then the order divides $2m$ but not m. Therefore its two-adic valuation is exactly $v_2(m)+1$. Conversely, factor identities such as $x^{2t}-1$ and $x^{2t+1}+1$ can construct a common divisor; an order obstruction by itself proves only necessity.

若奇质数 p 下 $a^m \equiv -1$，阶整除 $2m$ 却不整除 m，故其二赋值恰为 $v_2(m)+1$。反过来，可用 $x^{2t}-1$ 与 $x^{2t+1}+1$ 等分解恒等式构造公约数；单有阶的障碍只证明必要性。

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 1 / 方法 1 [NT04-M01] (continued / 续)

Practice / 配套练习: AIME-2001-II-Q10 (WB p. 519), AIME-2002-I-Q07 (WB p. 219), AIME-2004-II-Q10 (WB p. 736), AIME-2019-I-Q14 (WB p. 523), AIME-2021-II-Q09 (WB p. 520)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 2 / 方法 2 [NT04-M02]

Prove a cycle by returning to the full determining state

回到完整决定状态来证明周期

Procedure

1. Identify how much information determines the next term: one residue, a pair, or a longer consecutive tuple.

2. Compute successive full states until the initial or a later state repeats. A repeated output alone does not prove a cycle.

3. Separate any nonperiodic prefix, then count complete cycles and the final partial cycle.

4. For a polynomial sequence, prove a period algebraically by showing its difference at shifted indices vanishes modulo the base.

通用步骤

1. 确定下一项由多少信息决定：一个剩余、一对数，或更长连续元组。

2. 计算完整状态直到初始或后续状态重复；单个输出重复不能证明周期。

3. 分开非周期前缀，再数完整周期与最后不完整周期。

4. 多项式数列应通过证明平移下标后的差模底数为零来确认周期。

> Example. Let $u_0 = 0, u_1 = 1$ and $u_{n+2} \equiv u_{n+1} + u_n \pmod 2$. Find u_{100}.

> 例题：设 $u_0 = 0, u_1 = 1, u_{n+2} \equiv u_{n+1} + u_n \pmod 2$，求 u_{100}。

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 2 / 方法 2 [NT04-M02] (continued / 续)

Worked solution

1. The determining pairs are $(0,1),(1,1),(1,0),(0,1)$.

2. The first pair returns after three updates, so the terms repeat $0,1,1$.

3. Since $100 \equiv 1 \pmod 3$, the answer is $u_{100} = 1$.

例题解析

1. 决定状态对依次为 $(0,1),(1,1),(1,0),(0,1)$。

2. 初始状态对三次更新后返回，故数列重复 $0,1,1$。

3. 因 $100 \equiv 1 \pmod 3$，故 $u_{100} = 1$。

A recurrence of order k normally needs k consecutive residues as its state.

k 阶递推通常需要连续 k 个剩余组成状态。

Practice / 配套练习: AIME-2017-I-Q03 (WB p. 517), AIME-2018-II-Q02 (WB p. 518), AIME-2023-II-Q13 (WB p. 205), AIME-2023-II-Q15 (WB p. 525)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 3 / 方法 3 [NT04-M03]

Split nonunit prefixes and use CRT for a composite modulus

分离不可逆前缀，并对合数模使用中国剩余定理

Procedure

1. Factor the modulus into prime powers. If the base shares a prime factor, first find when its power becomes zero locally.

2. For the remaining unit parts, reduce exponents by a certified order or Euler period.

3. Combine local residues by CRT. For a sum of distinct residues, look for a fixed-point-free pairing such as $r \mapsto m - r$.

通用步骤

1. 将模数分为质数幂；底数若含相同质因数，先求其幂何时在该局部变为零。

2. 其余可逆部分，以确认的阶或欧拉周期约化指数。

3. 用中国剩余定理合并局部剩余。对不同剩余之和，可寻找 $r \mapsto m - r$ 等无不动点配对。

> Example. Find $6^{20} \pmod{40}$.

> 例题：求 6^{20} 模 40 的余数。

Worked solution

1. Modulo 8, every power from the third onward is zero, so $6^{20} \equiv 0$.

2. Modulo 5, the base is one, so $6^{20} \equiv 1$.

3. The unique residue modulo 40 that is zero modulo 8 and one modulo 5 is 16.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 3 / 方法 3 [NT04-M03] (continued / 续)

例题解析

1. 模 8 时，从三次幂起均为零，故 $6^{20} \equiv 0$。

2. 模 5 时底数为一，故 $6^{20} \equiv 1$。

3. 模 40 唯一满足模 8 余零、模 5 余一的剩余为 16。

Do not apply Euler to a base that is not coprime to the local modulus.

底数不与局部模数互质时不能使用欧拉定理。

For a power tower a^{b^N}, first find an exponent period h for the outer unit base and only then reduce b^N modulo h. If b is not coprime to h, split h into prime powers and handle its zero-producing local prefix before using any order.

对幂塔 a^{b^N}，先求外层单位底数的指数周期 h，再将 b^N 模 h 约化。若 b 不与 h 互质，须将 h 分为质数幂，先处理产生零的局部前缀，再使用乘法阶。

Practice / 配套练习: AIME-2011-I-Q11 (WB p. 521), AIME-2017-I-Q14 (WB p. 125), AIME-2025-II-Q13 (WB p. 217)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 4 / 方法 4 [NT04-M04]

Lift an order or a root to a prime-square modulus

把阶或根提升到质数平方模

Procedure

1. Find the root or exact order modulo the prime first.

2. For an order d, expand $(a^d)^t = (1+pc)^t$ modulo p^2; if $p \nmid c$, the order multiplies by p.

3. For a root a, write $a+pt$ and solve the resulting linear derivative congruence; inspect every initial root.

4. For higher powers use an explicitly justified lifting valuation, and keep singular derivative cases separate.

通用步骤

1. 先求模该质数的根或准确乘法阶。

2. 若模质数阶为 d，将 $(a^d)^t = (1+pc)^t$ 模 p^2 展开；若 $p \nmid c$，则阶乘以 p。

3. 对根 a 写成 $a+pt$，求所得导数线性同余，并检查每个初始根。

4. 更高次质数幂须使用已证明的赋值提升，并单列导数为零的奇异情形。

> Example. Find the order of 2 modulo 25.

> 例题：求 2 模 25 的乘法阶。

Worked solution

1. The order modulo 5 is 4, since $2^2 = -1$ modulo 5. Any order modulo 25 is a multiple of 4.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 4 / 方法 4 [NT04-M04] (continued / 续)

2. $2^{4t} = 16^t = (1+15)^t \equiv 1 + 15t \pmod{25}$.

3. This is one precisely when $5 \mid t$, proving the least exponent is 20.

例题解析

1. 模 5 的阶为 4，因 $2^2 = -1$；所以模 25 的阶必为 4 的倍数。

2. $2^{4t} = 16^t = (1+15)^t \equiv 1 + 15t \pmod{25}$。

3. 它等于一当且仅当 $5 \mid t$，故最小指数为 20。

A root may lift uniquely even though its order calculation uses a different argument.

根的唯一提升与阶的计算可能需要不同论证。

For odd p, $p \mid A - B$ and $p \nmid AB$ imply $v_p(A^t - B^t) = v_p(A - B) + v_p(t)$. The binomial expansion proves the rule first for $p \nmid t$, then for each additional factor of p.

Cubing on units modulo powers of 3 is singular: for units a, b with $a \equiv b \pmod 3$, $v_3(a^3 - b^3) = v_3(a - b) + 1$. Thus representatives modulo 3^k map injectively by cubing into the unit cube classes modulo 3^{k+1}, precisely the classes congruent to $\pm 1 \pmod 9$. The image has $2 \cdot 3^{k-1}$ elements, equal to the number of inputs.

对奇质数 p，若 $p \mid A - B, p \nmid AB$，则 $v_p(A^t - B^t) = v_p(A - B) + v_p(t)$。先用二项式展开证明 $p \nmid t$ 的情况，再逐次加入 p 因子即得。

模 3 幂的单位立方映射是奇异情形：若单位 a, b 满足 $a \equiv b \pmod 3$，则 $v_3(a^3 - b^3) = v_3(a - b) + 1$。故模 3^k 代表元的立方单射到模 3^{k+1} 的单位立方类，后者恰为模 9 余 ± 1 的类；其元素数 $2 \cdot 3^{k-1}$ 与输入数相同。

Section 3.4 Orders, modular powers & cycles

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Method 4 / 方法 4 [NT04-M04] (continued / 续)

Practice / 配套练习: AIME-2018-I-Q11 (WB p. 522), AIME-2020-I-Q12 (WB p. 524), AIME-2024-I-Q13 (WB p. 526), AIME-2025-I-Q15 (WB p. 527)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 5 / 方法 5 [NT04-M05]

Refine an exponent fixed point one prime-power layer at a time

逐层细化指数不动点的剩余

Procedure

1. Split the target congruence into coprime prime powers and obtain a first constraint on the exponent itself.

2. Use that constraint to reduce the power modulo a small prime power, obtaining a stronger residue class for the exponent.

3. Repeat only while the established exponent period justifies the next refinement, then recombine and verify the original equation.

通用步骤

1. 将目标同余拆为互质质数幂，并先得到指数本身的初步限制。

2. 用该限制约化小质数幂模下的幂，得到指数更强的剩余类。

3. 只有已知指数周期允许时才继续细化，最后合并并验证原方程。

> Example. Find all residues $n \pmod{20}$ satisfying $n \equiv 3^n \pmod{10}$.

> 例题：求满足 $n \equiv 3^n \pmod{10}$ 的全部模 20 剩余类。

Worked solution

1. The right side is odd, so n is odd. Powers of 3 have period four modulo 10.

2. If $n \equiv 1 \pmod{4}$, the equation requires $n \equiv 3 \pmod{10}$, giving $n \equiv 13 \pmod{20}$.

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Method 5 / 方法 5 [NT04-M05] (continued / 续)

3. If $n \equiv 3 \pmod 4$, it requires $n \equiv 7 \pmod{10}$, giving $n \equiv 7 \pmod{20}$. Both classes work, and the parity split exhausts all cases.

例题解析

1. 右侧为奇数，故 n 为奇数；3 的幂模 10 的周期为四。

2. 若 $n \equiv 1 \pmod 4$，原式要求 $n \equiv 3 \pmod{10}$，得 $n \equiv 13 \pmod{20}$。

3. 若 $n \equiv 3 \pmod 4$，则须 $n \equiv 7 \pmod{10}$，得 $n \equiv 7 \pmod{20}$。两类均成立，奇偶分类已穷尽。

Each stronger exponent class must follow from the previous one; guessing a long cycle is not a proof of minimality.

更强的指数剩余类须由前一步推出，猜测长周期不能证明最小性。

Practice / 配套练习: AIME-2021-II-Q13 (WB p. 516)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 6 / 方法 6 [NT04-M06]

Make a universal geometric sum vanish in every local case

使全称等比和在每种局部情形均消失

Procedure

1. Use multiplicativity to reduce a divisor-sum condition to $1+a+\cdots+a^N$ for prime-power factors.

2. Modulo a prime, separate $a=0$, $a=1$, and $a\neq 0,1$; the last case permits division by $a-1$.

3. For a least universal exponent, prove necessity with actual admissible inputs attaining the required residues or orders.

通用步骤

1. 用积性把约数和条件约为质数幂因子的 $1+a+\cdots+a^N$。

2. 模质数分别处理 $a=0, a=1, a\neq 0,1$；最后一种才可除以 $a-1$。

3. 求最小全称指数时，须用确实可取且达到所需剩余或阶的输入证明必要性。

> Example. Find the least positive N such that $a+a^2+\cdots+a^N\equiv 0 \pmod 5$ for every integer a.

> 例题：求最小正整数 N，使每个整数 a 都满足 $a+a^2+\cdots+a^N\equiv 0 \pmod 5$。

Worked solution

1. At $a=1$, necessity gives $5\mid N$. At $a=2$, the geometric formula gives $2^N=1$, so $4\mid N$.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 6 / 方法 6 [NT04-M06] (continued / 续)

2. Thus $20 \mid N$. For $N = 20$, $a = 0$ is automatic and $a = 1$ gives a multiple of 5.

3. For the other nonzero residues, $a^4 = 1$ and $a - 1$ is invertible, so $a(a^{20} - 1)/(a - 1) = 0$. The minimum is 20.

例题解析

1. $a = 1$ 给出 $5 \mid N$；$a = 2$ 时等比公式给出 $2^N = 1$，故 $4 \mid N$。

2. 故 $20 \mid N$。当 $N = 20$，$a = 0$ 自动成立，$a = 1$ 得 5 的倍数。

3. 其余非零剩余满足 $a^4 = 1$，且 $a - 1$ 可逆，故 $a(a^{20} - 1)/(a - 1) = 0$；最小值为 20。

A universal assertion needs all residue cases, including the one where the geometric denominator vanishes.

全称结论须覆盖所有剩余情形，包括等比公式分母为零者。

Practice / 配套练习: AIME-2021-I-Q14 (WB p. 612)